REFERENCE · FOR AGENT CODE AND AGENT GIT, EVERY PHASE

# Spareway — Engineering Reference

The exact names, shapes and numbers: TypeScript contracts, query keys, every database function, error codes and messages, Edge Function payloads, the sample data with the numbers tests must hit, test plan, git workflow, templates, demo runbook and known pitfalls.

| USE WITH | RULE | VERSION |
| --- | --- | --- |
| Architecture & Spec (what) · Design System (look) | Names here are final. Need a change → CONTRACT CHANGE REQUEST (12.4) | Final v6 · 29 Oct 2026 |

## 0. Contents


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## 1. Contracts — `src/types/domain.ts` (written in Phase 0)

Field lists below are the contract. Write them as TypeScript types/interfaces with exactly these names. DB rows use the generated `Database` types; these are the app-level shapes.

### 1.1 FeatureModule (every `features/*/module.tsx` default-exports one)

| Field | Type | Meaning |
|---|---|---|
| id | string | = folder name (`trucks`, `load-planner`…); also the i18n namespace |
| order | number | Menu order (Home 0, Trucks 10, Trips 20, Map 30, Loads 40, Earnings 50, Settings 99) |
| nav? | { labelKey, icon (lucide component), path, phone: 'tab' \| 'more' \| 'hidden' } | Rail + phone tab item |
| routes? | { path, element (lazy), titleKey }[] | Inside the shell, behind RequireAuth |
| publicRoutes? | { path, element (lazy), guard: 'publicOnly' \| 'none' }[] | No shell (auth pages, loader page) |
| slots? | Partial<Record<SlotName, SlotEntry[]>> | SlotEntry = { id, order, component, when?(props) → boolean, labelKey?, icon?, count?(props) } |

### 1.2 Slots

| SlotName | Host (owner) | Props | Filled by |
|---|---|---|---|
| home.actions | Home (0·Code) | — | 1·Code New trip · 1·Git Plan load · 2·Code Live map · 3·Git Find extra loads |
| home.widgets | Home (0·Code) | — | 3·Git Extra earnings available · 4·Git This week |
| settings.sections | Settings (0·Code) | — | 4·Code Save map for offline |
| truck.tabs | Truck page (1·Code) | truckId | 1·Git Load layout · 2·Code Location |
| trip.steps | Trip page (1·Code) | tripId | 1·Git Load plan (2) · 2·Code Loader (3) · 2·Git Route (4) · 3·Git Extra loads (5) · 4·Git Earnings (6) |
| trip.actions | Trip page (1·Code) | tripId | (small extra buttons; none planned) |
| cargo.addMethods | Cargo list (1·Code) | tripId, onAdded | 3·Code Scan with camera · Speak |
| loader.actions | Loader page (2·Code) | token, job | 3·Code Scan with camera · Speak |

SlotEntry = { id, order, component, labelKey?, icon?, when?(props) → boolean, done?(props) → boolean (for `trip.steps`: shows ✓), count?(props) }.

### 1.3 Errors, auth and language

| Type | Fields |
|---|---|
| AppError | code: ErrorCode (section 5) · message: string · cause?: unknown |
| AuthStatus | 'loading' \| 'signedOut' \| 'preparing' \| 'ready' \| 'error' |
| Language | useLang() → { lang: 'en' \| 'hi', toggle(), set(lang) } · t(key) for UI text · td(value) for stored demo values (dictionary 7.6) · tm(kind, meta) for alerts/activity (templates 7.7) |
| UseAuth | session · user · profile · company · settings · status · signInWithPassword(email, password) · signUp({fullName, companyName, email, password}) → { needsConfirmation } · signInWithGoogle(next?) · sendPasswordReset(email) · updatePassword(pw) · resendConfirmation(email) · signOut() · refreshProfile() |

### 1.4 Packing (F12)

| Type | Fields |
|---|---|
| PackContainer | lengthCm · widthCm · heightCm · payloadKg |
| PackItem | key (`<cargoItemId>#<n>`) · cargoItemId · label · l · w · h · weightKg · unloadOrder · loadOrder? · fragile · uprightOnly · stackable · maxTopLoadKg? |
| Placement | key · cargoItemId · label · x · y · z · l · w · h · orientation (0–5) · unloadOrder · weightKg · fragile · step · fixed? |

| Type | Fields |
|---|---|
| UnplacedReason | 'TOO_LARGE' \| 'OVER_PAYLOAD' \| 'NO_SPACE' \| 'NO_SUPPORT' \| 'ORDER_CONFLICT' |
| PackMetrics | volumeUsedPct · weightUsedPct · usedLengthCm · freeVolumeM3 · remainingPayloadKg · cogXPct · cogYPct · itemCount · unplacedCount |
| PackResult | engineVersion · inputHash · placements: Placement[] · unplaced: {key, label, reason}[] · metrics: PackMetrics · warnings: {code: 'COG_LONGITUDINAL'\|'COG_LATERAL'\|'HEAVY_HIGH', message}[] |
| PackOptions | supportRatio (default 1.0) · heavyBelow (true) · epsilon (0.5) |
| Stored in DB | `load_plans.placements` = Placement[] · `unplaced` · `metrics` = PackMetrics · `engine_version` · `input_hash`; `loading_progress.step_index` = Placement.step, `placement_key` = key |

### 1.5 Loader job (returned by `loader_get_job`)

```
LoaderJob = {
  company: string,
  trip:  { id, code, status, expiresAt },
  truck: { regNo, model, lengthCm, widthCm, heightCm, payloadKg },
  stops: [{ id, seq, type: 'origin'|'pickup'|'drop', name }],
  plan:  null | { id, version, placements: Placement[], metrics: PackMetrics },
  progress: [{ step, key, status: 'pending'|'done'|'skipped'|'issue', note }]
}   or   { error: 'LINK_INVALID' }
```

### 1.6 Routes (F15) and extra cargo (F17)

| Type | Fields |
|---|---|
| RouteVariant | variant: 'fastest'\|'shortest'\|'no_tolls' · distanceKm · durationMin · geometry (GeoJSON LineString) · restricted · cached |
| CostBreakdown | fuelInr · tollsInr · driverInr · waitingInr · totalInr · score · tollCount · checkpointCount |
| Halt | kind: 'break'\|'overnight'\|'no_entry_wait'\|'checkpoint' · name · lat · lng · atMin · durationMin · poiId? |
| Schedule | halts: Halt[] · stopEtas: {stopId, eta}[] · warnings: {code, message}[] |
| Feasibility | ok · reason?: 'OFF_CORRIDOR'\|'WRONG_DIRECTION'\|'TOO_LATE'\|'OVER_PAYLOAD'\|'NO_SPACE'\|'ORDER_CONFLICT'\|'LOW_MARGIN' · detail |
| ExtraCandidate | listingId · code · pickupKm · dropKm · detourKm · detourMin · cost {detourInr, tollsInr, handlingInr, waitingInr, delayInr, totalInr} · netBenefitInr · minViableRateInr · recommended · approximate · explanation {en, hi} · feasibility |

### 1.7 Smart input (F19, F20)

| Type | Fields |
|---|---|
| ParsedCargoItem | label · lengthCm · widthCm · heightCm · quantity · weightKg? · fragile · uprightOnly · stackable · confidence (0–1) · notes · source: 'camera'\|'voice' |
| VoiceParseResult | items: ParsedCargoItem[] · missing: ('length'\|'width'\|'height'\|'quantity'\|'weight')[] · unsure: boolean · transcript |
| Loader insert row (loader_add_cargo) | label · lengthCm · widthCm · heightCm · weightKg · quantity · dropStopId · fragile · uprightOnly · stackable · imagePath? · confidence? |

## 2. React Query keys (`src/lib/queryKeys.ts`) and invalidation

| Key | Data | Invalidated by |
|---|---|---|
| ['me'] | profile + company + settings | Settings save, reset sample data |
| ['home','stats'] | dashboard_stats | truck/trip/plan changes, realtime |
| ['alerts', {open}] | alerts | resolve, realtime alerts |
| ['activity'] | activity_log latest | realtime activity_log |
| ['fleetStats', period] | fleet_daily_stats | reset sample data |
| ['trucks'] · ['truck', id] | trucks | truck create/update, realtime trucks |
| ['trips', {status}] · ['trip', id] · ['trip', id, 'stops'] | trip_overview, trips, trip_stops | wizard, status change, accept_extra_load, route select (ETAs) |
| ['trip', id, 'cargo'] | cargo_items | add/edit/delete, camera/voice, realtime cargo_items |

| Key | Data | Invalidated by |
|---|---|---|
| ['trip', id, 'plans'] · ['plan', id, 'progress'] | load_plans, loading_progress | save/activate plan, realtime loading_progress |
| ['trip', id, 'routes'] | route_options | route plan / select |
| ['trip', id, 'extra'] | trip_extra_loads + candidates | accept, plan change |
| ['listings', {status}] | freight_listings | post, accept |
| ['pois', bbox] | pois | — |
| ['tripFinancials', id] | trip_financials | accept, route select |
| ['loaderJob', token] | loader_get_job (polling 5 s) | own writes (refetch immediately) |

staleTime 30 s · retry 1 (0 for 4xx) · sign-out clears the whole cache · realtime handler only invalidates (never writes cache by hand).

## 3. Database functions (RPC) — exact reference

| Function | Args | Returns | Errors | Caller |
|---|---|---|---|---|
| seed_sample_data() | — | { seeded: boolean } | NOT_SIGNED_IN | 0·Code bootstrap |
| reset_sample_data() | — | { reset: true } | NOT_SIGNED_IN | 0·Code Settings |
| next_trip_code() | — | text (TRP-2026-0032…) | — | 1·Code new trip |
| next_listing_code() | — | text (LD-2001…) | — | 3·Git post a load |
| activate_load_plan(p_plan_id) | uuid | void | PLAN_NOT_FOUND | 1·Git load plan |
| create_loader_link(p_trip_id) | uuid | { token, expires_at } | TRIP_NOT_FOUND | 2·Code loader step |
| disable_loader_link(p_trip_id) | uuid | void | — | 2·Code |
| loader_get_job(p_token) | text | LoaderJob (1.5) | { error: LINK_INVALID } | 2·Code loader page (anon) |
| loader_update_step(p_token, p_plan_id, p_step, p_status, p_note) | text, uuid, int, 'pending'\|'done'\|'skipped'\|'issue', text | { ok: true } | { error: LINK_INVALID \| BAD_STATUS \| PLAN_CHANGED } | 2·Code loader page (anon); issue → owner alert `loader_issue` with meta {code, step, note}; first write sets trip to loading |
| loader_add_cargo(p_token, p_items, p_source) | text, jsonb array (1–50 rows, 1.7), 'camera'\|'voice' | { ok: true, added } | { error: LINK_INVALID \| BAD_SOURCE \| BAD_ITEMS \| BAD_DROP_STOP } | 3·Code loader page (anon); owner alert `loader_cargo` with meta {code, count, source} |
| accept_extra_load(p_extra_load_id, p_pickup_after_seq, p_drop_after_seq) | uuid, int, int | { pickup_stop_id, drop_stop_id, pickup_seq, drop_seq } | NOT_SIGNED_IN · EXTRA_LOAD_NOT_FOUND · ALREADY_ACCEPTED · DROP_BEFORE_PICKUP · LISTING_NOT_FOUND · LISTING_NOT_OPEN | 3·Git accept; reuses a stop within 2 km |
| km_between(lat1, lng1, lat2, lng2) | double ×4 | double km | — | SQL only |
| loader_trip_for_token(p_token) | text | (trip_id, company_id) | — | ai-cargo (service role only) |

Raised errors arrive in supabase-js as `error.message` containing the code; map them to `AppError` by searching for the code string. JSON errors (`{error}`) come back as data — check `data.error` first.

## 4. Views (all filtered to the caller's company by RLS)

| View | Columns |
|---|---|
| trip_overview | id, company_id, code, status, planned_departure, actual_departure, base_freight_inr, truck_id, reg_no, model, payload_kg, truck_volume_m3, origin_name, final_stop_name, stop_count, cargo_weight_kg, cargo_volume_m3, cargo_units, active_plan_id, plan_volume_used_pct, plan_weight_used_pct, route_distance_km, route_duration_min, route_total_cost_inr |
| dashboard_stats | active_trucks, total_trucks, drivers_on_duty, total_drivers, live_trips, avg_utilisation_pct, on_time_pct |

| View | Columns |
|---|---|
| trip_financials | trip_id, company_id, code, status, planned_departure, base_freight_inr, extra_revenue_inr, extra_cost_inr, route_cost_inr, distance_km, net_earnings_inr (= base + extra revenue − route cost − extra cost) |

## 5. Error codes and messages (use these exact texts in i18n `common:errors.*`)

| Code | Where | English | हिंदी |
|---|---|---|---|
| INVALID_LOGIN | login | Email or password is wrong | ईमेल या पासवर्ड ग़लत है |
| EMAIL_NOT_CONFIRMED | login | Please confirm your email first | पहले अपना ईमेल कन्फ़र्म करें |
| EMAIL_TAKEN | signup | An account with this email already exists | इस ईमेल से खाता पहले से है |
| WEAK_PASSWORD | signup / reset | Use at least 8 characters with a letter and a number | कम से कम 8 अक्षर, एक अक्षर और एक अंक |
| RATE_LIMITED | auth | Too many tries — please wait a minute | बहुत बार कोशिश हुई — एक मिनट रुकें |
| NETWORK | any | No connection — check your internet | इंटरनेट नहीं है — कनेक्शन देखें |
| NOT_SIGNED_IN | RPC | Please log in again | कृपया दोबारा लॉग इन करें |
| LINK_INVALID | loader | This loading link has expired or was turned off. Ask the owner for a new one. | यह लोडिंग लिंक ख़त्म हो गया या बंद है। मालिक से नया लिंक माँगें। |
| PLAN_CHANGED | loader | The plan was updated | प्लान बदल गया है |
| BAD_DROP_STOP | loader cargo | Choose where this cargo is dropped | यह माल कहाँ उतरेगा, चुनें |
| TRIP_NOT_FOUND · PLAN_NOT_FOUND | owner | This trip or plan no longer exists | यह ट्रिप या प्लान अब नहीं है |
| LISTING_NOT_OPEN · ALREADY_ACCEPTED | extra cargo | This load was just taken — refreshing | यह लोड अभी किसी ने ले लिया — रीफ़्रेश हो रहा है |
| DROP_BEFORE_PICKUP | extra cargo | Drop must come after pickup | उतराई लदाई के बाद होनी चाहिए |
| ORS_ERROR · ORS_QUOTA | routes | Routes are unavailable right now — showing saved routes | अभी रूट नहीं मिल रहे — सेव रूट दिखा रहे हैं |
| AI_TIMEOUT · AI_PARSE | camera/voice | Couldn't read that — try again or type it | समझ नहीं आया — दोबारा करें या लिखें |
| TOO_LARGE (upload) | camera | Photo too large — retake it | फ़ोटो बहुत बड़ी है — फिर से लें |
| AUTH_REQUIRED | Edge Functions | Please log in again | कृपया दोबारा लॉग इन करें |

| Reason (not an error) | Message pattern (en) |
|---|---|
| TOO_LARGE (packing) | “Too big for this truck in any position (95 × 260 × 90 cm)” |
| OVER_PAYLOAD | “Too heavy: 16 t > 12.4 t left” |
| NO_SPACE | “No space left — 0.6 m³ needed, 0.2 m³ free” |
| NO_SUPPORT | “Nothing strong enough to stack on” |
| ORDER_CONFLICT | “Would block goods that leave earlier” |
| OFF_CORRIDOR · WRONG_DIRECTION | “38 km off your route” · “Goes the opposite way” |
| TOO_LATE | “Can't reach Bhopal before Sat 6:00 pm” |
| LOW_MARGIN | “Earns only ₹120 after costs — needs ≥ ₹1,840” |

## 6. Edge Function payloads (examples; the contract is Architecture section 9)

### 6.1 route-plan

```
POST /functions/v1/route-plan      Authorization: Bearer <owner JWT>
{ "tripId": "…", "coordinates": [[76.9366,28.3540],[75.8306,26.7787],[72.5947,22.9536],[73.0483,19.2813]],
  "variants": ["fastest","shortest","no_tolls"],
  "truck": { "lengthM": 9.75, "widthM": 2.43, "heightM": 2.43, "weightT": 31, "axleloadT": 10 } }
→ { "ok": true, "data": { "routes": [ { "variant": "fastest", "distanceKm": …, "durationMin": …,
       "geometry": { "type": "LineString", "coordinates": [...] }, "restricted": true, "cached": false } ] } }
→ { "ok": false, "error": { "code": "ORS_QUOTA", "message": "…" } }
```

### 6.2 ai-cargo

```
POST /functions/v1/ai-cargo     (owner: Authorization Bearer <JWT>;  loader: no JWT, loaderToken in body)
{ "mode": "image", "tripId": "…", "loaderToken": null,
  "images": [ { "base64": "…", "mime": "image/jpeg" } ],
  "reference": { "name": "A4 sheet", "lengthCm": 29.7, "widthCm": 21 }, "lang": "hi" }
{ "mode": "voice", "loaderToken": "3f9c…", "transcript": "दस बाई दस का बॉक्स …", "lang": "hi" }
→ { "ok": true, "data": { "scanId": "…", "model": "…", "imagePaths": ["<company>/<trip>/<uuid>.jpg"],
     "items": [ { "label": "Carton", "lengthCm": 60, "widthCm": 40, "heightCm": 45, "quantity": 16,
                  "fragile": false, "uprightOnly": false, "stackable": true, "confidence": 0.82,
                  "notes": "", "countsPerView": [16, 15] } ], "warnings": ["COUNT_MISMATCH"] } }
```

**Agent Git** mocks these with fixtures in `__fixtures__/` ; **Agent Code** deploys the functions and runs the same examples against the real endpoint.

## 7. Sample data — what every new company gets (from `_seed_company`)

### 7.1 Trucks

| Reg no | Model | Inside L×W×H cm | Payload kg | Axles | ₹/km | Status | Driver | Fuel % |
|---|---|---|---|---|---|---|---|---|
| MP09 HG 4821 | Tata Signa 3118.T · 32 ft MXL container | 975 × 243 × 243 | 15,000 | 3 | 34 | active | Ramesh Yadav | 72 |
| RJ14 GC 7735 | Ashok Leyland 2820 · 24 ft closed | 732 × 244 × 244 | 10,000 | 3 | 28 | idle | Suresh Meena | 88 |
| GJ01 KT 2208 | Eicher Pro 3015 · 19 ft closed | 579 × 214 × 214 | 9,000 | 2 | 22 | active | Vikram Patel | 54 |
| MH04 LQ 9913 | BharatBenz 1617R · 22 ft container | 670 × 244 × 244 | 9,000 | 2 | 25 | maintenance | Anil Pawar | 40 |
| MP09 KR 1147 | Tata 1412 LPT · 17 ft closed | 518 × 213 × 213 | 7,000 | 2 | 20 | offline | Mahesh Chouhan | 12 |
| RJ14 TB 3360 | Tata Ultra 1918.T · 20 ft closed | 610 × 229 × 229 | 9,000 | 2 | 24 | active | Imran Khan | 61 |

### 7.2 Trips

| Code | Truck | Status | Stops | Base freight | Plan |
|---|---|---|---|---|---|
| TRP-2026-0031 (demo) | MP09 HG 4821 | planned, departs tomorrow 06:00 IST | 0 Manesar warehouse → 1 Jaipur (Sitapura) → 2 Ahmedabad (Narol) → 3 Mumbai (Bhiwandi), with windows | ₹62,000 | none (the demo plans it) |
| TRP-2026-0029 | GJ01 KT 2208 | in_transit (left 3 h ago) | Ahmedabad (Aslali) ✓ → Vadodara (Makarpura) ✓ → Surat (Sachin GIDC) | ₹21,000 | active seed plan: 78.4% vol / 61.2% wt |
| TRP-2026-0030 | RJ14 TB 3360 | completed (on time) | Jaipur (VKIA) ✓ → Himmatnagar ✓ | ₹28,500 | active seed plan: 64.0% / 60.0% |

### 7.3 Demo cargo (TRP-2026-0031) — totals 104 units · 2,638 kg · ≈ 19.8 m³

| Item | Client | L×W×H cm | kg/unit | Qty | Drop | Flags |
|---|---|---|---|---|---|---|
| Split AC indoor unit 1.5T | Shree Electronics, Jaipur | 95 × 32 × 38 | 14 | 24 | 1 Jaipur | upright |
| Ceiling fan (carton of 4) | Shree Electronics, Jaipur | 60 × 40 × 45 | 18 | 16 | 1 Jaipur | priority low |
| Split AC outdoor unit 1.5T | Patel Appliances, Ahmedabad | 90 × 40 × 70 | 38 | 20 | 2 Ahmedabad | upright |
| LED TV 43 inch | Patel Appliances, Ahmedabad | 105 × 18 × 68 | 11 | 12 | 2 Ahmedabad | fragile, upright, not stackable, high |
| Washing machine front-load | Konkan Retail, Bhiwandi | 65 × 65 × 90 | 72 | 8 | 3 Mumbai | upright |
| Refrigerator 260 L | Konkan Retail, Bhiwandi | 62 × 70 × 165 | 52 | 6 | 3 Mumbai | upright, not stackable, high |
| Microwave oven 25 L | Konkan Retail, Bhiwandi | 52 × 42 × 35 | 13 | 18 | 3 Mumbai | fragile |

Truck volume 975 × 243 × 243 cm = 57.6 m³ → booked volume ≈ 34%, weight 2,638 / 15,000 ≈ 18% (manifest bars). Packed plan reference: 104 placed, 0 unplaced, used length ≈ 640 cm (well under 975).

### 7.4 Marketplace samples (private to each company) and expected outcome for TRP-2026-0031

| Code | Goods · shipper | Pickup → drop | Qty × size cm · kg/unit | Offer ₹ | Expected |
|---|---|---|---|---|---|
| LD-1041 | Marble tiles · Kishangarh Marble House | Kishangarh → Ahmedabad (Narol) | 8 × 120×60×40 · 180 | 16,000 | candidate (fragile) |

| Code | Goods · shipper | Pickup → drop | Qty × size cm · kg/unit | Offer ₹ | Expected |
|---|---|---|---|---|---|
| LD-1042 | Textile bales · Beawar Textiles | Beawar → Mumbai (Bhiwandi) | 20 × 100×60×50 · 90 | 24,000 | candidate |
| LD-1043 | Handicraft cartons · Udaipur Co-op | Udaipur → Vadodara | 25 × 60×50×50 · 20 | 9,000 | candidate |
| LD-1044 | Sanitaryware · Himmatnagar Ceramics | Himmatnagar → Mumbai (Bhiwandi) | 30 × 70×55×45 · 25 | 15,500 | candidate |
| LD-1045 | Auto parts · Vadodara Auto Components | Vadodara → Mumbai (Bhiwandi) | 18 × 80×60×60 · 45 | 14,000 | candidate (demo accept) |
| LD-1046 | Fabric rolls · Surat Silk Mills | Surat (Kamrej) → Mumbai (Bhiwandi) | 24 × 150×40×40 · 35 | 9,500 | candidate |
| LD-1047 | Packaged snacks · Bharuch Foods | Bharuch → Surat (Kamrej) | 40 × 60×40×40 · 6 | 4,200 | candidate |
| LD-1048 | Paper reels · Vapi Paper Products | Vapi → Mumbai (Bhiwandi) | 3 × 110×110×120 · 600 | 11,000 | candidate |
| LD-1049 | Soybean bags · Malwa Agro | Indore → Bhopal | 40 × 80×50×30 · 50 | 18,000 | OFF_CORRIDOR |
| LD-1050 | Office furniture · Mumbai Movers | Mumbai (Bhiwandi) → Ahmedabad | 10 × 120×80×90 · 60 | 20,000 | WRONG_DIRECTION |
| LD-1051 | Steel coils · Gujarat Steel Traders | Vadodara → Mumbai (Bhiwandi) | 4 × 150×150×120 · 4,000 | 45,000 | OVER_PAYLOAD (16 t) |
| LD-1052 | Pottery · Jaipur Blue Pottery | Jaipur (Sitapura) → Udaipur | 6 × 50×40×40 · 15 | 300 | LOW_MARGIN |

Test rule: ≥ 3 of LD-1041…1048 recommended (exact net benefit depends on settings and route; assert reasons and ordering, not rupee values).

### 7.5 Alerts, activity, stats and the dashboard numbers

| Item | Value |
|---|---|
| Alerts (4) | Engine warning MP09 KR 1147 (critical) · Critical fuel level MP09 KR 1147 12% (critical) · Route deviation RJ14 TB 3360 (warning) · Mumbai no-entry window TRP-2026-0031 (info) |
| Activity (6) | trip departed Aslali · delivered Vadodara · fuelled Himmatnagar 180 L · trip completed Himmatnagar · geofence alert resolved · trip planned Manesar → Mumbai |
| fleet_daily_stats | 30 days, all `is_sample`; extra-load revenue only on the last 6 days |
| Dashboard stats (fresh company) | Active 3/6 · Drivers on duty 4/6 · Live trips 1 · Avg utilisation 71.2% · On-time 100% |

### 7.6 Hindi dictionary for demo data (`src/i18n/data/hi.json`, written in Phase 0)

Keys are the exact stored English values; `td(value)` returns the Hindi when Hindi is on. Truck numbers, trip/load codes and numbers are never translated. Vansh and Bhavya proof-read this list before Day 1.

| English (stored) | हिंदी |
|---|---|
| Manesar warehouse | मानेसर गोदाम |
| Gurugram | गुरुग्राम |
| Jaipur (Sitapura) | जयपुर (सीतापुरा) |
| Jaipur | जयपुर |
| Ahmedabad (Narol) | अहमदाबाद (नारोल) |
| Ahmedabad | अहमदाबाद |
| Mumbai (Bhiwandi) | मुंबई (भिवंडी) |
| Bhiwandi | भिवंडी |
| Ahmedabad (Aslali) | अहमदाबाद (असलाली) |
| Vadodara (Makarpura) | वडोदरा (मकरपुरा) |

| English (stored) | हिंदी |
|---|---|
| Vadodara | वडोदरा |
| Surat (Sachin GIDC) | सूरत (सचिन GIDC) |
| Jaipur (VKIA) | जयपुर (VKIA) |
| Himmatnagar | हिम्मतनगर |
| Kishangarh | किशनगढ़ |
| Beawar | ब्यावर |
| Udaipur | उदयपुर |
| Surat (Kamrej) | सूरत (कामरेज) |
| Bharuch | भरूच |
| Vapi | वापी |
| Indore | इंदौर |
| Bhopal | भोपाल |
| Mumbai | मुंबई |

| English (stored) | हिंदी |
|---|---|
| Tata Signa 3118.T · 32 ft MXL container | टाटा सिग्ना 3118.T · 32 फ़ुट MXL कंटेनर |
| Ashok Leyland 2820 · 24 ft closed body | अशोक लेलैंड 2820 · 24 फ़ुट बंद बॉडी |
| Eicher Pro 3015 · 19 ft closed body | आयशर प्रो 3015 · 19 फ़ुट बंद बॉडी |
| BharatBenz 1617R · 22 ft container | भारतबेंज़ 1617R · 22 फ़ुट कंटेनर |
| Tata 1412 LPT · 17 ft closed body | टाटा 1412 LPT · 17 फ़ुट बंद बॉडी |
| Tata Ultra 1918.T · 20 ft closed body | टाटा अल्ट्रा 1918.T · 20 फ़ुट बंद बॉडी |
| Ramesh Yadav | रमेश यादव |
| Suresh Meena | सुरेश मीणा |
| Vikram Patel | विक्रम पटेल |
| Anil Pawar | अनिल पवार |
| Mahesh Chouhan | महेश चौहान |
| Imran Khan | इमरान ख़ान |
| Split AC indoor unit 1.5T | स्प्लिट AC इनडोर यूनिट 1.5 टन |
| Ceiling fan (carton of 4) | छत का पंखा (4 का डिब्बा) |
| Split AC outdoor unit 1.5T | स्प्लिट AC आउटडोर यूनिट 1.5 टन |
| LED TV 43 inch | LED टीवी 43 इंच |
| Washing machine front-load | वॉशिंग मशीन फ़्रंट-लोड |
| Refrigerator 260 L | फ़्रिज 260 लीटर |
| Microwave oven 25 L | माइक्रोवेव ओवन 25 लीटर |
| Pharma cartons | दवाइयों के डिब्बे |
| Marble tiles crate | मार्बल टाइल्स का क्रेट |
| Shree Electronics, Jaipur | श्री इलेक्ट्रॉनिक्स, जयपुर |
| Patel Appliances, Ahmedabad | पटेल अप्लायंसेज़, अहमदाबाद |
| Konkan Retail, Bhiwandi | कोंकण रिटेल, भिवंडी |
| Zydus distributor | ज़ायडस वितरक |
| Sanghvi Ceramics | संघवी सिरेमिक्स |
| Marble tiles | मार्बल टाइल्स |
| Textile bales | कपड़े की गाँठें |
| Handicraft cartons | हस्तशिल्प के डिब्बे |

| English (stored) | हिंदी |
|---|---|
| Ceramic sanitaryware | सिरेमिक सैनिटरीवेयर |
| Auto parts | ऑटो पार्ट्स |
| Fabric rolls | कपड़े के रोल |
| Packaged snacks | पैक्ड नमकीन |
| Paper reels | काग़ज़ के रोल |
| Soybean bags | सोयाबीन की बोरियाँ |
| Office furniture | ऑफ़िस फ़र्नीचर |
| Steel coils | स्टील कॉइल |
| Pottery | मिट्टी के बर्तन |
| Kishangarh Marble House | किशनगढ़ मार्बल हाउस |
| Beawar Textiles | ब्यावर टेक्सटाइल्स |
| Udaipur Handicrafts Co-op | उदयपुर हस्तशिल्प सहकारी |
| Himmatnagar Ceramics | हिम्मतनगर सिरेमिक्स |
| Vadodara Auto Components | वडोदरा ऑटो कंपोनेंट्स |
| Surat Silk Mills | सूरत सिल्क मिल्स |
| Bharuch Foods | भरूच फ़ूड्स |
| Vapi Paper Products | वापी पेपर प्रोडक्ट्स |
| Malwa Agro | मालवा एग्रो |
| Mumbai Movers | मुंबई मूवर्स |
| Gujarat Steel Traders | गुजरात स्टील ट्रेडर्स |
| Jaipur Blue Pottery | जयपुर ब्लू पॉटरी |
| Polished tiles in wooden crates | लकड़ी के क्रेट में पॉलिश टाइल्स |
| Compressed cotton bales | दबी हुई कपास की गाँठें |
| Wooden decor, packed | पैक की हुई लकड़ी की सजावट |
| Boxed wash basins | डिब्बों में वॉश बेसिन |
| Brake assemblies in cartons | डिब्बों में ब्रेक असेंबली |
| Synthetic fabric rolls, wrapped | लिपटे हुए सिंथेटिक कपड़े के रोल |
| Light cartons, keep dry | हल्के डिब्बे, सूखा रखें |
| Kraft paper reels on pallets | पैलेट पर क्राफ़्ट पेपर रोल |
| Off corridor (Indore → Bhopal) | रूट से बाहर (इंदौर → भोपाल) |
| Wrong direction (Mumbai → Ahmedabad) | उल्टी दिशा (मुंबई → अहमदाबाद) |
| Too heavy for remaining payload | बचे वज़न के लिए बहुत भारी |
| Low offer: likely negative net benefit | कम भाड़ा: शायद घाटा |
| Consumer durables for three distributors. Demo trip. | तीन वितरकों के लिए घरेलू सामान। डेमो ट्रिप। |

| English (stored) | हिंदी |
|---|---|
| Route deviation | रूट से भटका |
| RJ14 TB 3360 left the planned route on NH48 near Himmatnagar. Driver rerouted via SH. | RJ14 TB 3360 हिम्मतनगर के पास NH48 पर तय रूट से हटा। ड्राइवर ने SH से रास्ता बदला। |
| Engine warning | इंजन चेतावनी |
| MP09 KR 1147 reported a diagnostic warning. Inspection required before next trip. | MP09 KR 1147 में इंजन की चेतावनी आई। अगली ट्रिप से पहले जाँच ज़रूरी। |
| Critical fuel level | ईंधन बहुत कम |
| MP09 KR 1147 fuel at 12%. Vehicle offline — refuel before dispatch. | MP09 KR 1147 में ईंधन 12% है। गाड़ी बंद है — भेजने से पहले ईंधन भरवाएँ। |

| English (stored) | हिंदी |
|---|---|
| Mumbai no-entry window | मुंबई नो-एंट्री समय |
| TRP-2026-0031 may reach Mumbai inside the 17:00–21:00 no-entry window. Check the route plan. | TRP-2026-0031 मुंबई में 17:00–21:00 नो-एंट्री समय में पहुँच सकती है। रूट प्लान देखें। |
| Trip departed Ahmedabad (Aslali) | ट्रिप अहमदाबाद (असलाली) से निकली |
| Delivered at Vadodara (Makarpura) | वडोदरा (मकरपुरा) में माल पहुँचा |
| Fuelled at Himmatnagar — 180 L | हिम्मतनगर में ईंधन भरा — 180 लीटर |
| Trip completed at Himmatnagar | हिम्मतनगर में ट्रिप पूरी |
| Geofence alert resolved | जियोफ़ेंस अलर्ट हल हुआ |
| Trip planned: Manesar → Jaipur → Ahmedabad → Mumbai | ट्रिप प्लान हुई: मानेसर → जयपुर → अहमदाबाद → मुंबई |

### 7.7 Message templates for problems and activity (by `kind` + `meta`)

The feature that writes an alert or activity row MUST fill `meta` with the keys below. Place names inside meta go through `td()`. Seeded rows have no meta and use the dictionary in 7.6.

| kind | English | हिंदी |
|---|---|---|
| loader_issue (alert) · meta {code, step, note} | Loader reported a problem — {code} · step {step}: {note} | लोडर ने समस्या बताई — {code} · स्टेप {step}: {note} |
| loader_cargo (alert) · meta {code, count, source} | Loader added cargo — {code}: {count} items by {source}. Re-plan the load. | लोडर ने माल जोड़ा — {code}: {source} से {count} सामान। लोड दोबारा प्लान करें। |
| overload (alert, 1·Code) · meta {code, kg, payloadKg} | Too heavy — {code}: {kg} of {payloadKg} | वज़न ज़्यादा — {code}: {kg} / {payloadKg} |
| plan_unplaced (alert, 1·Git) · meta {code, count} | {count} items didn't fit in {code} | {code} में {count} सामान नहीं आया |
| city_entry (alert, 2·Git) · meta {code, city, from, to, wait} | {code} reaches {city} in no-entry hours ({from}–{to}) — wait {wait} | {code} {city} में नो-एंट्री समय ({from}–{to}) में पहुँचेगी — {wait} रुकें |
| plan_activated (activity) · meta {version, volumeUsedPct} | Load plan v{version} in use — {volumeUsedPct}% space used | लोड प्लान v{version} चालू — {volumeUsedPct}% जगह भरी |
| extra_load_accepted (activity) · meta {listing_code, trip_code, pickup, drop} | Extra load {listing_code} added to {trip_code}: {pickup} → {drop} | अतिरिक्त लोड {listing_code} {trip_code} में जोड़ा: {pickup} → {drop} |
| trip_created (activity, 1·Code) · meta {code} | New trip {code} | नई ट्रिप {code} |
| truck_status (activity, 1·Code) · meta {reg, status} | {reg} is now {status} | {reg} अब {status} |
| source words | camera · voice | कैमरा · आवाज़ |

### 7.8 Earnings test fixture (F23)

`net = base_freight + Σ accepted offers − route total cost − Σ incremental costs`. Fixture for unit tests (not real data): base 62,000; route cost 40,000; one accepted offer 14,000 with incremental cost 3,550 → net 32,450; before extra loads 22,000; gain +10,450.

## 8. Test plan

| Command | Runs |
|---|---|
| npm run typecheck | tsc --noEmit |
| npm run lint | eslint |
| npm run test | vitest run (jsdom) |
| npm run check | all three — must pass before every push |
| npm run build | production build — must pass without .env.local |

| Feature | Must-have unit tests |
|---|---|
| F1/F4 | module discovery + slot ordering · nav hidden for missing features · build without env · POI validator |
| F2 | auth error mapping · guards redirect · signup states (mocked) |
| F3 | format (₹ en-IN, km, kg/t, ft, IST) · units ft⇄cm · en/hi key parity · td() on every 7.6 value · tm() for every 7.7 kind in both languages |
| F5/F6 | problem sorting · resolve invalidates · settings schema limits |
| F7–F9 | ft⇄cm schema · unloadOrderForStops · cargo totals (104 / 2,638 kg) |
| F10 | 8 cubes fill 100% · validatePlan invariants · determinism · fitAdditional keeps fixed · demo trip 104/0 < 1.5 s · over-payload fixture |
| F11/F13 | usePlanForTrip (mocked worker/api) · positionWords · loader polling pause/resume · PLAN_CHANGED |
| F14–F16 | route-plan contract example · chainage · tolls · planSchedule (break, overnight, no-entry wait) · costRoute · position along a line |
| F17/F18 | validator · ≥ 25 voice sentences (Devanagari digits, डेढ़/ढाई/सवा/साढ़े, बाय/by/x, इंच, नाज़ुक) |
| F19/F20 | corridor + direction · timing · weight · space · profit · LD-1049…1052 reasons · ≥ 3 recommended · post schema |
| F21–F23 | offline queue order · earnings fixture 7.8 |

| Manual QA (Agent Code) | Devices / modes |
|---|---|
| Every phase PDF Verify list + every “Code must verify” list | Laptop Chrome 1280 px · Android Chrome over hotspot (https LAN) · Vercel URL (from Phase 4·Code) |
| Language and modes | Tap the Hindi button on every screen (labels + demo data) · low-data on/off · reduced motion |
| Accounts | Two owner accounts (company isolation, public listings) · loader link in a private tab |

## 9. Performance budgets

| Item | Budget |
|---|---|
| First load (4G, mid phone) | < 3 s to interactive on the dashboard; initial JS < 350 KB gzip (3D, map, charts lazy-loaded) |
| Loader page | < 150 KB gzip JS, no three.js, no map |
| Packing | 104 items < 1.5 s in the worker; UI never blocks |
| 3D | ≥ 30 fps on a mid phone with 104 boxes; dpr ≤ 1.5 |
| Polling / realtime | loader 5 s (10 s low-data), pause when hidden; realtime channels ≤ 5 per page |
| Images | ≤ 1280 px, ≤ 700 KB each, max 3 per scan |

## 10. Translations (i18n)

- Namespaces: `common` (app/i18n) + one per feature (= module id). Files `features/<id>/i18n/en.json` and `hi.json`, loaded with the module.
- Keys: `section.key` in camelCase (`trucks:form.payloadHelp`); plurals with i18next `_one` / `_other`; interpolation `{{count}}`.
- Parity test fails the build if a key exists in one language only.
- Language: profile `preferred_lang` → browser (`hi*` → Hindi) → English. Loader page: device-saved choice, default Hindi.

- Wording: Design System section 10 glossary — use those words exactly.
- Demo data: `src/i18n/data/hi.json` from 7.6 (exact English keys). Alerts/activity: `common:templates.<kind>` from 7.7.
- Hindi button: `useLang().toggle()`; saves to `profiles.preferred_lang` (owner) or localStorage (loader page).

## 11. Git workflow (Phase 0 alone, then Code local + Git cloud)

| Phase part | Agent | Branch | Starts from |
|---|---|---|---|
| 0·Code | Code | `main` | Repository with the docs committed |
| 1·Code | Code | `code/phase1-trucks-trips` | Phase 0 (tag v0) |
| 1·Git | Git | `git/phase1-load-plan` | Phase 0 (tag v0) |
| 2·Code | Code | `code/phase2-loader-map` | Merge 1 (tag v1) |
| 2·Git | Git | `git/phase2-routes` | Merge 1 (tag v1) |
| 3·Code | Code | `code/phase3-camera-voice` | Merge 2 (tag v2) |
| 3·Git | Git | `git/phase3-extra-loads` | Merge 2 (tag v2) |
| 4·Code | Code | `code/phase4-offline-deploy` | Merge 3 (tag v3) |
| 4·Git | Git | `git/phase4-earnings` | Merge 3 (tag v3) |

1. Phase 0 (Agent Code alone) commits straight to `main` and ends with tag `v0`.
2. From Phase 1, both parts of a phase start from the same `main` commit. Agent Git: in claude.ai/code pick repo `spareway`, branch `main`, environment `spareway-cloud`; tell it the branch name to create.
3. Commits: `feat(scope): …`, `fix(scope): …`, `test(scope): …`, `chore: …`; scope = feature id. Small and often; push at least every hour (cloud sessions can expire).
4. Never push to `main`, never force-push, never rewrite another agent's commits.
5. Merge (Agent Code, fresh session): `git checkout main && git pull` → `git merge --no-ff origin/code/…` → check → `git merge --no-ff origin/git/…` → apply Git's SQL files with MCP → regenerate `database.ts` → `npm run check && npm run build` → run Git's "Code must verify" list → push main → tag.
6. Conflicts: `database.ts` → regenerate, never hand-merge; `package-lock.json` → take main's and `npm install`; any other file → the owner from Architecture 4.4 wins, the other side re-applies its change.
7. If an agent's session dies: start a new one with "Continue Phase N·X from docs/PROGRESS.md on branch …".

## 12. Templates

### 12.1 CLAUDE.md (Phase 1·Code writes it; ≤ 120 lines)

```
# Spareway — rules for Claude Code
Who am I?  Local app with Supabase MCP = Agent Code.  Cloud (claude.ai/code, no .env.local) = Agent Git.
Read first: docs/Spareway_01_Plan_Architecture_and_Spec.pdf, docs/Spareway_02_Design_System.pdf,
            docs/Spareway_03_Engineering_Reference.pdf, docs/phases/<your phase>.pdf, docs/PROGRESS.md
Ask first, plan, then build.  Never guess names (DB types, these docs).  Stay in your folders (Architecture 4.4).
Design rules: Design System section 1 (simple UI) + 6 (Hindi button).  Contracts: Engineering Reference 1.
Every string en + hi; demo values via td(); alerts/activity via tm(kind, meta).
Commands: npm run dev | check | build.   Done = check + build pass, Verify list, PROGRESS updated, pushed.
```

### 12.2 PROGRESS.md section (one per phase part)

```
## Phase 1·Git — Packing engine, 3D load plan, map style   (branch git/phase1-load-plan)   status: ready to merge
Built: …                          How to demo (clicks): …
Stubs / fixtures left: …          Contract change requests: none | see below
Code must verify (Git only):      [ ] plan TRP-2026-0031 on the real DB  [ ] activate → loader page shows steps …
Notes for the other agent: …
```

### 12.3 DECISIONS.md row

```
| 2026-10-31 | 2·Git | Support ratio 1.0 by default | user answer Q1; overhang caused unstable stacks |
```

### 12.4 CONTRACT CHANGE REQUEST (write in PROGRESS.md, tell the user)

```
CONTRACT CHANGE REQUEST — from 3·Git
File/contract: src/types/domain.ts ExtraCandidate      Change: add field `approximate: boolean`
Why: …      Workaround used meanwhile: local type in extra-loads/types.ts      Owner to apply: Agent Code at merge
```

### 12.5 Bug report (team → Agresh)

```
Screen: Trip detail › Cargo   Device: Android Chrome, Hindi   Steps: 1… 2… 3…
Expected: …   Happened: …   Screenshot/video: …   Severity: blocks demo | annoying | cosmetic
```

## 13. Definition of done (every phase part)

☐ Questions asked and answered; plan approved.
☐ Only owned folders changed (check `git diff --stat main`).
☐ `npm run check` + `npm run build` pass (Git: also without .env.local — always true in the cloud).
☐ Phase Verify list done (Code) or tests + “Code must verify” list written (Git).
☐ Design PDF section 16 checklist done.
☐ en + hi complete; honesty badges in place; no dead buttons.
☐ PROGRESS.md + DECISIONS.md updated; branch pushed; user told “ready to merge” + top 3 things to check.

## 14. Demo-day runbook and the 5-minute demo

| Before judging | Check |
|---|---|
| Accounts | A fresh demo owner account signed up on the Vercel URL the evening before (demo fleet seeded); a second account for the marketplace |
| Devices | Laptop on the hotspot; phone with the loader link already opened once (certificate accepted); charger |
| Fallbacks | Routes cached for TRP-2026-0031 · typed voice ready · manual cargo form · backup video (Yash) · Reset sample data in Settings |
| Email | If confirmation emails are slow, turn Confirm email OFF in Supabase before judges sign up |

| Time | Show | Say (one line) |
|---|---|---|
| 0:00 | Home (fresh account) → tap हिंदी → tap English | “A small truck owner's day — in his own language with one tap.” |
| 0:30 | TRP-2026-0031 Cargo tab | “104 units booked, truck only a third full by volume.” |
| 1:00 | Plan load → 3D → Play sequence | “Stop order, stacking and fragile rules — Mumbai deepest, Jaipur by the doors.” |
| 1:45 | Send to loader → phone scans QR → Done, Problem | “The loader needs no app or login — Hindi steps on his phone.” |
| 2:30 | Route tab → three routes → Select | “Full trip cost with tolls, halts and Mumbai no-entry hours.” |
| 3:15 | Extra cargo → rejects with reasons → Accept LD-1045 | “Only loads that make money after detour, tolls and waiting.” |
| 4:00 | Speak Hindi on the loader phone → owner alert | “दस बाई दस का बॉक्स… — cargo by voice, a person confirms.” |
| 4:30 | Earnings tab → before vs after | “More space used, more earned per trip.” (the three hero numbers) |

## 15. Known pitfalls and fixes

| Symptom | Cause → fix |
|---|---|
| Tailwind classes do nothing | Tailwind v4 has no tailwind.config by default: use `@import "tailwindcss"` + `@theme` in index.css and the `@tailwindcss/vite` plugin |
| shadcn looks white/zinc | Map shadcn CSS variables to our tokens in index.css (Design 4.6) |
| Map is blank / unstyled | Import `maplibre-gl/dist/maplibre-gl.css`; use `react-map-gl/maplibre`; style URL must be same-origin or CORS-enabled |
| R3F errors about React | Match @react-three/fiber major to the installed React (v9 ↔ React 19); check node_modules versions |
| Worker import fails in build | `new Worker(new URL('./worker.ts', import.meta.url), { type: 'module' })` + Comlink wrap |
| Camera/mic blocked on phone | Needs HTTPS: use the basic-ssl https LAN URL or Vercel; accept the certificate once |
| Speech recognition missing | Only Chrome/Edge (webkitSpeechRecognition); show typed fallback elsewhere |
| Empty data, no error | RLS: row's company_id is not the user's — never set company_id manually; check you're signed in |
| Realtime events never arrive | Table must be in the supabase_realtime publication (done in SQL Part 1); subscribe after sign-in |
| Loader page 401/permission | Loaders must only call the loader_* RPCs; never select tables with the anon client |
| Google login returns to localhost on the phone | Add the current origin to Supabase Redirect URLs; redirectTo uses window.location.origin |
| Cloud build fails on env | env.ts must read env lazily (runtime screen), never throw at import time |
| Cloud agent tries the Supabase MCP | The MCP must be added at local scope (not in a committed .mcp.json) |
| Hindi button leaves English bits | A string not in i18n, or a stored demo value not in data/hi.json, or an alert written without meta — fix at the source |
| Cloud session can't install packages | Environment network must be Trusted/Custom with defaults; setup script `npm ci` |

| Symptom | Cause → fix |
|---|---|
| PWA serves stale app | Update toast → reload; during dev the service worker is disabled |

## 16. Trucking glossary

| Term | Meaning |
|---|---|
| MXL / SXL | Multi-axle / single-axle long container trucks (32 ft class) |
| Payload | Weight the truck may carry (gross − empty) |
| Part load (PTL) / full load (FTL) | Cargo that uses part of / the whole truck |
| Backhaul / return load | Cargo for the way back or along the way |
| Dhaba | Highway restaurant where drivers rest |
| No-entry hours | City times when heavy trucks may not enter |
| Checkpoint | State border / RTO check that adds delay |
| NH48 | National Highway Delhi–Jaipur–Ahmedabad–Mumbai |
| Chainage | Distance along the route from the start (km) |
| Detour | Extra km to reach a pickup/drop off the route |
| COG | Centre of gravity of the load |
| LIFO | Last in, first out — goods unloaded first are loaded last, by the doors |